Combinational Logic Circuits - Adders & Subtractors

1. The Half Adder

Theory: A half adder is a combinational circuit that adds two single-bit binary numbers (A and B). It has two outputs: the **Sum (S)** and the **Carry (C)**. It is called a "half" adder because it cannot accept a carry-in from a previous addition.

Truth Table:

Inputs		Outputs	
A	B	S (Sum)	C (Carry)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

K-Map Simplification for Half Adder

For Sum (S):

Notice the checkerboard pattern. The `1`s cannot be grouped, which is the classic signature of an XOR gate.

$A \backslash B$	0	1
0	0	**1**
1	**1**	0

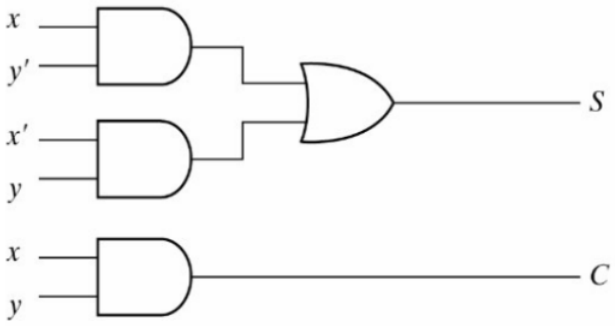
- **Sum Equation:** $S = \bar{A}B + A\bar{B} = \mathbf{A \oplus B}$

For Carry (C):

$A \backslash B$	0	1
0	0	0
1	0	**1**

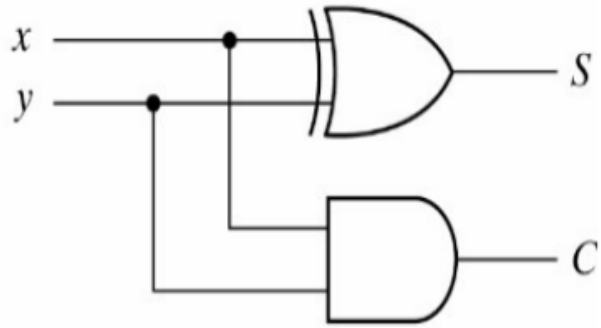
- **Carry Equation:** $C = \mathbf{AB}$

Circuit Diagram for Half Adder



S=XY'+X'Y
C=XY

OR



S = X ⊕ Y
C=XY

2. The Full Adder

Theory:

A full adder improves upon the half adder by adding three single-bit numbers: A, B, and a **Carry-In (C_{in})** from a previous less-significant bit stage. This is the building block needed to add multi-bit binary numbers.

Truth Table:

Inputs			Outputs	
A	B	C_{in}	S (Sum)	C_{out} (Carry Out)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-Map Simplification for Full Adder

For Sum (S):

Again, a perfect 3-variable checkerboard. No groupings are possible.

$A \backslash BC_{in}$	00	01	11	10
0	0	**1**	0	**1**
1	**1**	0	**1**	0

- **Sum Equation:** $S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in} = \mathbf{A \oplus B \oplus C_{in}}$

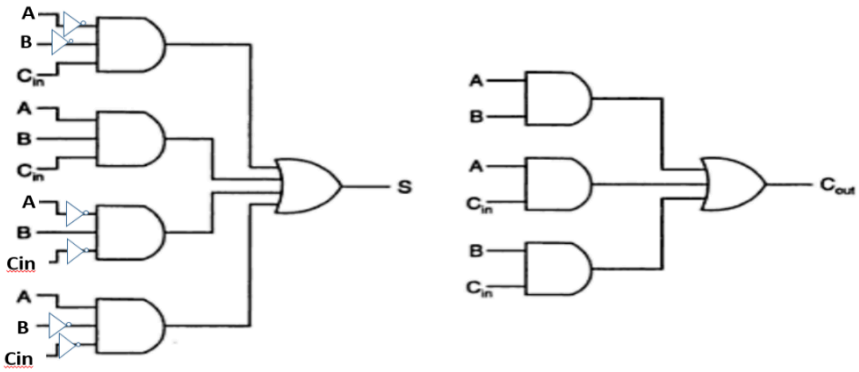
For Carry-Out (C_{out}):

Here we can form three separate groups of two.

$A \backslash BC_{in}$	00	01	11	10
0	0	0	**1**	0
1	0	**1**	**1**	**1**

- **Carry-Out Equation:** $C_{out} = \mathbf{AB + BC_{in} + AC_{in}}$

Circuit Diagram for Full Adder



$S = A'B'C_{in} + ABC_{in} + AB'C_{in} + A'BC_{in}$

$C = BC_{in} + AB_{in} + AC_{in}$

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3. The Half Subtractor

Theory:

A half subtractor performs $A - B$. It produces a **Difference (D)** and a **Borrow (B_{out})**. The borrow is triggered (set to 1) only when you try to subtract 1 from 0.

Truth Table:

Inputs		Outputs	
A	B	D (Difference)	B_{out} (Borrow)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

K-Map Simplification for Half Subtractor

For Difference (D):

Identical to the Half Adder Sum!

$A \backslash B$	0	1
0	0	**1**
1	**1**	0

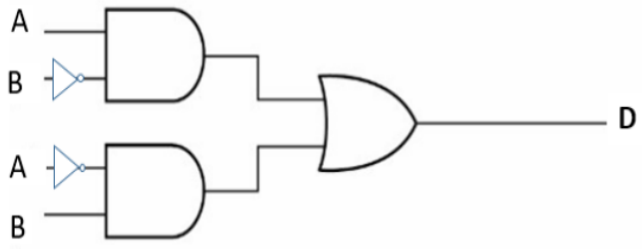
- **Difference Equation:** $D = \bar{A}B + A\bar{B} = \mathbf{A \oplus B}$

For Borrow (B_{out}):

$A \backslash B$	0	1
0	0	**1**
1	0	0

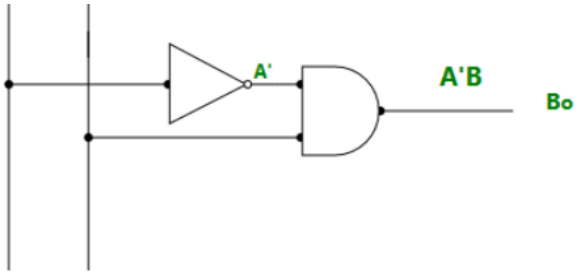
- **Borrow Equation:** $B_{out} = \mathbf{\bar{A}B}$

Circuit Diagram for Half Subtractor



OR

A B



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4. The Full Subtractor

Theory:

A full subtractor performs $(A - B - B_{in})$, taking into account a **Borrow-In (B_{in})** requested by a previous, less-significant stage.

Truth Table:

Inputs			Outputs	
A	B	B_{in}	D (Difference)	B_{out} (Borrow Out)
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

K-Map Simplification for Full Subtractor

For Difference (D):

Identical to the Full Adder Sum!

$A\backslash BB_{in}$	00	01	11	10
0	0	**1**	0	**1**
1	**1**	0	**1**	0

- **Difference Equation:** $D = \mathbf{A} \oplus \mathbf{B} \oplus \mathbf{B_{in}}$

For Borrow-Out (B_{out}):

$A\backslash BB_{in}$	00	01	11	10
0	0	**1**	**1**	**1**
1	0	0	**1**	0

- **Borrow-Out Equation:** $B_{out} = \mathbf{\bar{A}B} + \mathbf{\bar{A}B_{in}} + \mathbf{BB_{in}}$

Circuit Diagram for Full Subtractor

